

HOW IT WORKS

One message, six seconds, seven steps

This page follows a single real message from the moment it arrives to the moment the answer appears. The timings, the thoughts and the tool results below are transcribed from a recorded run, not tidied up for the page.

A

Ankor Care online



Party TOMORROW and this stupid thing just stopped!!! It says E05

- Reading your photo robot vacuum
- Reading your message anxious · deadline tomorrow
- Checking what E05 means brush roll blocked

Party tomorrow, so let's go fast. E05 means the brush roll is blocked.

1. Turn the robot over.
2. Cut away any hair around the brush.
3. Refit the brush until it clicks.

Still E05 after that? Say so and I'll get a person on it.

Recorded run · answered in 6.1s

THE MESSAGE

"I'm hosting a party TOMORROW and this stupid thing just stopped!!!! It says E05"

with a photograph of the machine



the customer's own photo

WHAT HAPPENS NEXT



01



Reads the photo

A picture is a fact, not decoration.

robot vacuum · brand not readable

0.9s

02



Reads the person

Anger, worry and a deadline are three different problems.

anxious · party tomorrow · says E05

1.2s

03



Works out the device

Guessing the model is worse than asking for the sticker.

kind known, exact model not yet

0.2s

04



Goes and looks it up

Thinks, calls one tool, reads the answer, decides.

E05 = brush roll blocked

2.8s

05



Writes the reply

Lead with the fix. Sympathy is a clause, not a paragraph.

3 steps · 1 source cited

1.6s

06



Checks itself

Six rules read the draft before the customer does.

passed, nothing rewritten

0.1s

THE ANSWER, 6.9 SECONDS LATER



"Party tomorrow, so let's go fast. E05 means the brush roll is blocked.

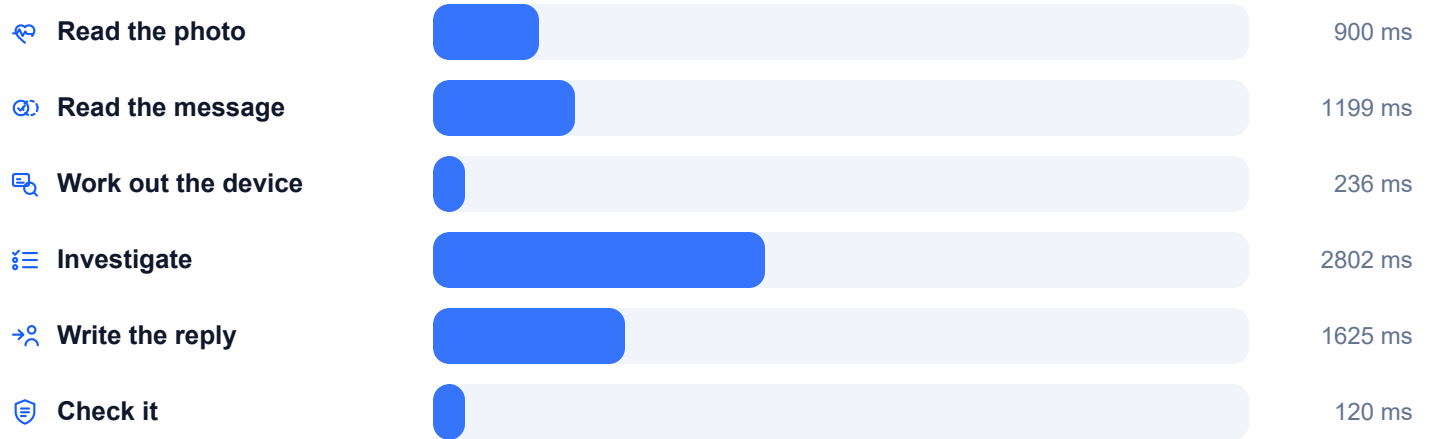
1. Turn the robot over.
 2. Cut away any hair wrapped around the brush.
 3. Refit the brush until it clicks.
- Still E05 after that? Tell me and I'll get a person on it."

No invented model number. No promise of a technician. One source, cited.

The whole turn, drawn. Every fact in the blue chips is what that step actually returned.

01 What happens in those six seconds

Six steps run before a single word reaches the customer. Bars are to scale; the numbers are the measured milliseconds of the recorded run.



Total 6.9 s. The customer sees the first words before the end of it: the reply streams while the last checks run.

02 Step by step, with what each one found

Each step is given one job and hands on a fact, not an opinion. That is what makes the trail readable an hour later.

01 Read the photo

900 ms

It is asked for facts, never opinions: the brand exactly as printed, the model, anything showing on a screen, visible damage, and where the model sticker lives on this kind of device.

robot_vacuum · brand unknown · no model visible · "look on the sticker underneath"

02 Read the message

1199 ms

How they feel, how strongly, whether there is a real deadline, what they actually want, which products and codes they named, and whether something has already been tried and failed.

anxious 0.8 · deadline “party tomorrow” · intent troubleshoot · code E05 · language en

03 Work out the device

236 ms

Five signals, cheapest first: the names people use, what this customer has bought, the words in the message, then the photo. Asking comes last, and it is asked once.

kind known (robot vacuum), exact model not, so it answers generally and asks for the sticker

04 Investigate

2802 ms

Think one sentence, look one thing up, read what came back, decide whether anything else is needed. It stops the moment it can help.

2 passes · lookup_error_code(E05) → “brush roll blocked” (249 ms) · enough to answer

05 Write the reply

1625 ms

Acknowledge in a clause, never a paragraph. Lead with the fix. Say nothing the lookups did not return.

574 characters · “acknowledge first, deadline noted, no upsell” · one citation

06 Check it

120 ms

Six rules read the draft before the customer does. They can rewrite it, remove a source that points at nothing, or send it back if it started talking about warranty without checking.

6 rules checked · passed, nothing rewritten

02b What “reading the person” actually produces

One step, one small object. Everything after it is shaped by these few fields, and none of it is guesswork about what the customer meant.

```
{
  "emotion": "anxious",           // how they sound
  "intensity": 0.8,               // how strongly
  "urgency": { "has_deadline": true, "deadline_hint": "party tomorrow" },
  "intent": "troubleshoot",      // what they want
  "entities": { "error_codes": ["E05"], "product_mentions": [] },
  "fix_failed": false,           // have they already tried something?
  "language": "en"               // answer in this
}
```

“anxious 0.8” with a deadline is why the reply opens with six words and then the fix. A calm customer with no deadline gets a different shape of answer from the same facts.

03 The loop inside step four

“Investigate” is not one call. It thinks, calls one tool, reads the result, and decides whether it can answer yet. In the recorded run it took two passes and one tool.

PASS 1 · THOUGHT

“Let me look up what error E05 means on your device.”

→ calls `lookup_error_code(code=E05)`

PASS 1 · RESULT

code recognised: Brush roll blocked · 249 ms

→ deterministic lookup, no model involved

PASS 2 · THOUGHT

“Good news, E05 is just a jammed brush roll, and it's a quick fix you can do tonight.”

→ decides: answer

The loop is capped, a tool that returned nothing counts as information, and the same tool is never called twice with the same arguments. The thought in each pass is written for the customer to read, and it is what appears in the live step trail.

04 “My S1 Pro isn't sucking”, which S1 Pro?

A breast pump and a robot vacuum share that name. Asking straight away is easy and wrong; the customer already told you more than they realise.

- 1 The alias table: “S1 Pro”, “omni s1”, “my robovac”
- 2 What this customer has actually bought
- 3 The words they used: “mopping” is not a breast pump
- 4 The photo: kind of device, brand, model sticker
- 5 Ask, once, with pictures of the candidates

Two rules learned the hard way

A generic phrase never pins a specific model. “My Anker power bank” used to resolve to one SKU because the alias table had exactly one match, and three turns later the agent was still talking about it.

And once the customer has answered, the answer holds. Every later turn starts from the device the conversation already settled, unless the customer plainly moves to another kind of device.

05 Warranty is not the model's opinion

The model chooses which record to look at. The record supplies the facts. A deterministic engine returns one of seven verdicts, and the reply may only phrase it.

 **covered**
In term, on file

 **covered_via_dealer**
Dealer invoice verified

 **covered_pending_verification**
Receipt supplied, seller to check

 **needs_proof**
Marketplace, no receipt yet

 **expired**
Out of term

 **not_covered_policy**
Damage, wear, unauthorised seller

 **escalate_human**
Nothing identifies the purchase

The inputs come from the records in a fixed order: this turn's order or dealer lookup, then the purchase this conversation already established, then the signed-in customer's own order for that product. Anything the model typed only fills gaps, and any disagreement is logged.

06 What it carries to the next message

The single-message tests passed all along. Every serious defect found in testing happened between turns, so what survives a turn is the design.

[resolved_sku](#)

The device they settled on

So “is it still under warranty?” never re-opens the picker they just answered.

[purchase](#)

The purchase the records established

So the second warranty question gets the same verdict as the first.

[failed_attempts](#)

Fixes that did not work

Counted across the conversation, not the turn. That is what makes escalation land on time.

[ticket](#)

The open ticket and its ETA

One ticket per conversation, so “how long will that take?” has an answer.

[safety](#)

A safety case, once raised

“Can I keep using it until the replacement comes?” after a swelling battery is still a safety question.

07 The tools it can reach for

Eight of the twelve. Each one comes back with a record, and the reply may only say what a record says.

<code>lookup_error_code</code>	What a code means, matched by shape so E05 = E-05 = E5, within the right kind of device.
<code>search_kb</code>	Manuals, FAQs and guides, narrowed to the customer's own model.
<code>get_troubleshooting_flow</code>	A curated, ordered repair flow for a known symptom.
<code>search_tickets</code>	Past tickets that were actually resolved, by symptom.
<code>lookup_order</code>	The customer's own order. Identity comes from the session, never from an argument.
<code>lookup_dealer_order</code>	A reseller invoice, matched against the dealer directory, authorisation included.
<code>check_warranty</code>	The deterministic engine. Reads the records, returns one of seven verdicts.
<code>create_ticket</code>	Hands over with the case written up: photo, steps tried, deadline.

08 What stops it making things up

Six rules read the draft before the customer does, in code, not by asking the model nicely.

G1

No coverage claim without the engine

G2

Safety outranks everything

If the draft drifts into warranty language and the engine never ran, the rule runs it and the draft is rebuilt around the verdict.

Swelling, smoke, a burning smell: stop using it, and the case goes to a person immediately.

G3

No invented specifics

A price, a SKU or a part name that appears in no tool result is removed before the customer sees it.

G4–G6

Tone, citations, and the deadline

Citations that point at nothing are stripped; a stated deadline must be acknowledged; the register has to match the emotion that was read.

09 When it cannot help

The interesting behaviour of a support agent is what it does when the answer is not available.

Nothing in the manuals

It says so and asks what the customer can see, rather than dressing six weak passages up as an answer.

The fix keeps failing

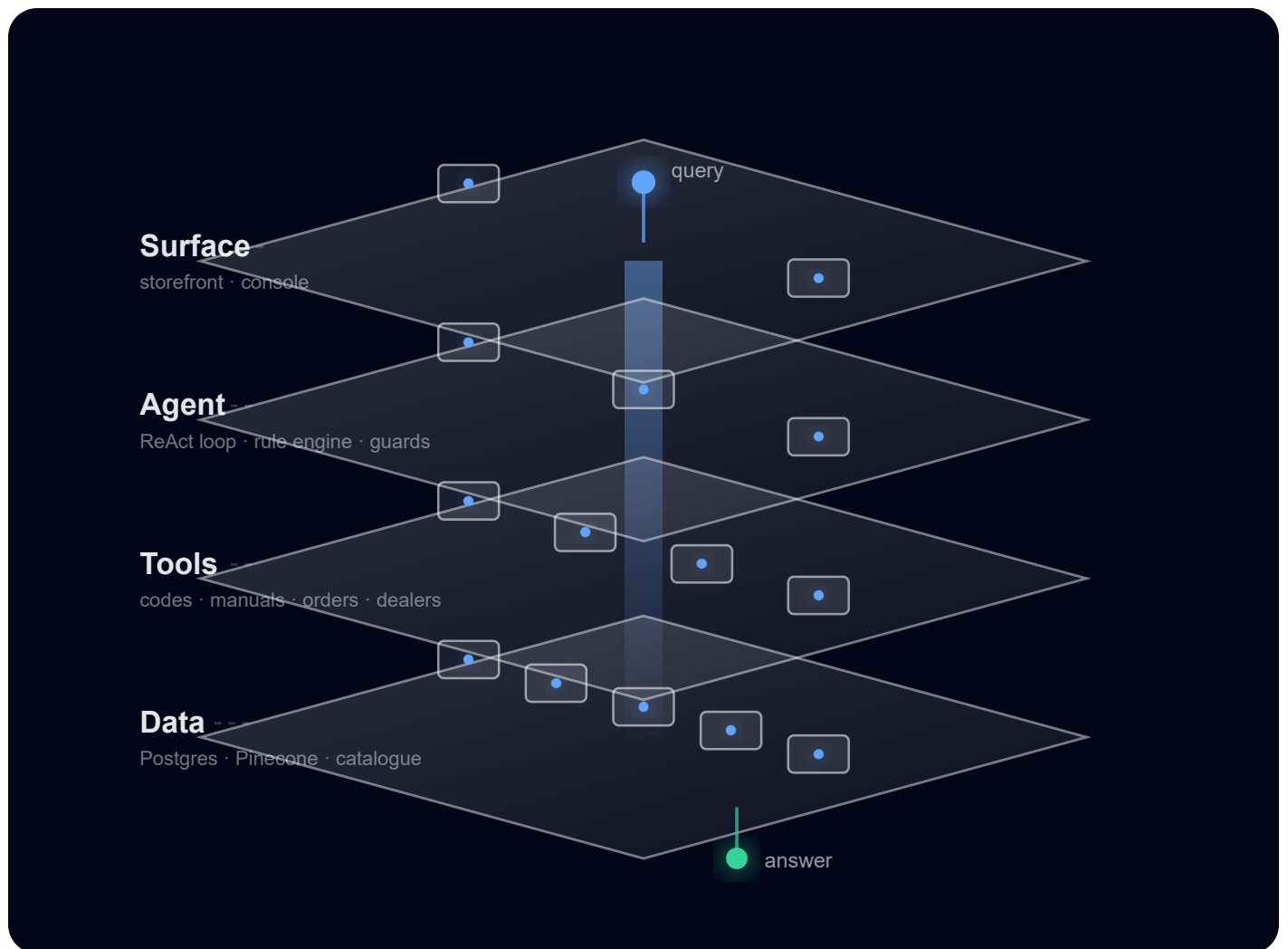
Failures are counted across the conversation. At the limit for that emotion it stops troubleshooting, opens one ticket and offers a person.

Not our product

A photo showing another brand is called out first, kindly, and it offers what it can still do.

10 What it is made of

Four layers. The surface the customer touches, the loop that decides, the tools that fetch records, and the stores those records live in. Nothing in a layer may invent what the layer below did not return.



Surface [Next.js](#)

A storefront with a launcher for the customer, a console with the full trail for the operator. Both read one conversation.

Agent [FastAPI](#) · [SSE](#)

A hand-written ReAct loop, a deterministic warranty engine beside it, six guard rules over the draft. Every stage, thought and tool result is streamed as it happens.

Tools [12](#)

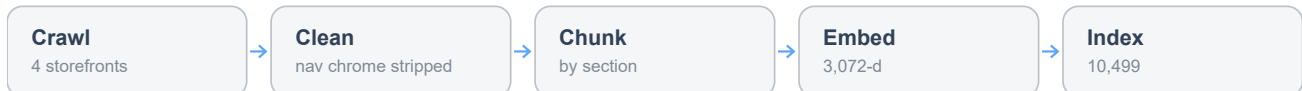
Error codes, manuals, repair flows, resolved tickets, orders, dealer invoices, the warranty engine, ticket creation. Each returns records, never prose.

Data [Postgres](#) · [Pinecone](#)

1,491 products with prices checked against the live stores, their manuals and error codes, orders and a dealer directory, and 10,499 vectors over the help corpus.

11 Where the knowledge came from

Nothing here is a fixture. The catalogue was crawled from the brands' own stores, the help corpus from their support sites, and both were checked back against the live pages.



Most of a support site is not support

Of 11.9k imported help rows, only about 510 were real articles; the rest was navigation furniture repeated on every page. Embedding it would have buried the manuals under menus.

Duplicates that look like variants

Products repeat across regional stores. They are linked rather than deleted, because a UK listing and a US listing are the same device at two prices.

Prices checked against the store

Every row with a URL was fetched and compared: 32 prices had drifted, 4 pages were gone, 40 products sat in the wrong category. All corrected, with a backup and an undo.

12 Which models, and why

Chosen by measurement on this workload, not by reputation. One support turn makes six to eight model calls, so latency per call is the whole budget.

Understanding, planning,
writing

[deepseek-chat](#)

Measured here: perception in 0.9s against 3.2s,
and warmer replies. The non-reasoning mode,

		because thinking tokens are latency the customer feels.
Reading photographs	deepseek-flash	The reasoning mode, only here. A photo is one call per turn, and it is the one that must not guess a brand.
Retrieval	gemini-embedding-001	3,072 dimensions, batched 100 chunks per request. HyDE rewrites the customer's words into manual language before the search runs.
Judging the agent	gpt-5	Deliberately a different family from the agent, so nothing grades its own style. It reads the whole transcript together with what every tool returned.

13 How we knew it worked

The single-message suite passed 27 of 27 from the beginning, and could not see a single one of the defects below. Every one of them happened between turns.

The harness

30 scripted customers, 64 turns, over the real HTTP API
It clicks the real buttons and sends real photographs
Chinese, Indonesian and English in the same suite
Per-turn structural checks, then a whole-transcript judge
246 unit tests, 33 written from the defects this found

12 → 29

conversations passing, out of 30

1.12 → 4.75

judge score on rule-following, 0-5

What it caught

- 01 Anyone could read any customer's orders by naming their email
- 02 The same dealer order got three different verdicts in three turns
- 03 An unauthorised grey-market seller came out covered
- 04 The product the customer had just picked was forgotten next turn

- 05 E05 on the screen never matched E-05 in the table
- 06 Still not working, said three times, got the same steps again
- 07 A safety warning did not survive to the next message
- 08 A second ticket opened, and the first one forgotten

All eight are fixed, and each has a regression test with the defect written into it.

[Ankor Care](#) · [How it works](#) · [Brochure](#) · [Poster](#)

Trace recorded from a live run of this build. Photo: [ajay_suresh](#), CC BY 2.0.